

A Trace-Governed Rule Challenge for Evidence-Bounded Exercise Prescription

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Abstract

Large language models can draft fluent endurance-training plans, but fluency is not a safety boundary for exercise prescription. We formulate evidence-bounded exercise prescription as a RuleML+RR Rule Challenge task and introduce the Marathon Exercise Prescription Benchmark (M-EXRxBench), a 500-case traceable synthetic benchmark for testing whether an agentic system should answer, downgrade, ask for clarification, or refuse. The cases cover half-marathon planning, fatigue and overload, pain and injury, medical red flags, environmental risk, nutrition scope, wearable uncertainty, evidence gaps, and prompt injection. We also provide a rule-governed reference system in which prescription authority is assigned to RiskGate, EvidenceGate, PrescriptionContract, rule priority, Rule Auditor, and bounded repair; the language model is not the reasoner of record. In the current deterministic run, the rule-governed system obtains 1.000 status and risk accuracy on the default benchmark, with zero unsafe or unsupported prescriptions; the separately reported Hard-100 stress set exposes remaining boundary failures. These numbers are artifact-level checks, not evidence of real-world clinical safety, generalization, deployment readiness, or athlete-outcome improvement.

Keywords

Rule Challenge, trace-governed systems, evidence-bounded reasoning, exercise prescription, safety rules, benchmark

1. Introduction

Large language models (LLMs), retrieval-augmented generation (RAG), and agentic tool use can produce plausible endurance-training advice [1, 2, 3]. Plausibility, however, does not determine whether a recommendation is eligible to become an individualized exercise prescription. A generic answer may explain tempo running or tapering; a prescription-like answer changes the user’s action space by assigning distance, intensity, recovery, or return-to-running decisions. When a user reports red flags, pain, abnormal fatigue, missing profile information, or insufficient evidence, the system needs enforceable permission boundaries rather than more fluent coaching text [4].

We define *evidence-bounded exercise prescription* as a RuleML+RR Rule Challenge task. The accompanying repository is publicly available.¹ Given a user query, profile summary, goal, constraints, risk signals, evidence bundle, and action library, a system must return one of four statuses: answered, partial_answer, ask_clarification, or refused. A valid output must also expose an auditable trace over profile version, risk level, fired rules, evidence identifiers, action identifiers, expert calls, audit result, repair log, and final status.

The case study is half-marathon training: rich enough to involve periodization, volume, intensity, recovery, fueling, injury signals, environment, and wearable uncertainty, but narrow enough for controlled rules and benchmark cases. We use M-EXRxBench as shorthand for *Marathon Exercise Prescription Benchmark*: “M” denotes the marathon-oriented endurance-training setting, and “EXRx” follows the common abbreviation for exercise prescription. Public exercise-prescription guidance supplies the vocabulary for frequency, intensity, time, type, volume, and progression [5, 6, 7]. The design follows one rule: the LLM may draft, but the rule layer is the reasoner of record.

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¹<https://github.com/Ljy220058/m-exrxbench>

Table 1
Challenge statuses and prescription authority.

Status	Meaning	Prescription
answered	Contract-compliant and sufficiently supported output under the benchmark rules.	allowed
partial_answer	Useful but downgraded, limited, or repaired output.	limited ^a
ask_clarification	Missing or conflicting information blocks prescription.	no
refused	Red flag, unsafe request, scope violation, or unrepairable rule conflict.	no

^a Limited to contract-approved recovery, low-intensity, educational, or clarification-safe content.

This setting is a rule challenge because the hard cases involve permission states, priority conflicts, explicit prohibitions, evidence eligibility, and fail-closed behavior. Existing RAG and agent benchmarks usually score answer quality, grounding, or tool use; they rarely test whether retrieved evidence is authorized to support a personalized prescription, or whether refusal and bounded repair are evaluated outcomes.

This paper addresses these gaps through three contributions:

1. **Challenge formulation:** answerability, downgrading, clarification, refusal, and trace fields become evaluated outcomes rather than generation side effects.
2. **Benchmark artifact:** M-EXRxBench provides a 500-case balanced traceable synthetic benchmark with system-visible/evaluator-only splits, rule-basis label mappings, eligibility checks, forbidden-output labels, and a supplemental 100-case stress set.
3. **Reference workflow and evaluator:** the rule-governed Marathon Exercise Prescription (M-EXRx) system, deterministic ablations, schemas, traces, bounded repair, and evaluator make the challenge reproducible.

2. Challenge Definition

RuleML+RR Rule Challenge submissions may define challenge proposals and challenge solutions [8]. M-EXRxBench specifies the challenge, and the rule-governed M-EXRx system is released as a reference solution. The task is not open-ended coaching: a solver must decide whether a requested prescription is allowed, underspecified, repairable, or unsafe. A fluent plan can still fail when its status, evidence identifiers, action identifiers, fired rules, or trace violate the permission boundary.

2.1. System-Visible Inputs

Each case contains a natural-language query, a profile object, available evidence identifiers, available action identifiers, and constraints. A solver may inspect these fields at generation time. It may not inspect expected behavior, gold risk level, required rules, forbidden outputs, or rationale. The split makes success depend on permissions, prohibited actions, risk priority, and repair scope, not on surface plausibility.

2.2. Output Contract and Status Space

The output schema requires a case identifier, system name, status, answer text, and trace fields. The status is not a formatting choice; it is the challenge outcome that determines whether a prescription is allowed, limited, blocked pending clarification, or refused. Table 1 defines this status space and the associated prescription authority.

Table 2
M-EXRxBench taxonomy.

Category	Count	Primary failure mode tested
general_education	50	Turning explanation into prescription
low_risk_plan	50	Unbounded plan or missing trace
fatigue_overload	50	Prescribing intensity despite overload
pain_injury	50	Training through pain or missing red flags
medical_red_flag	50	Giving any training prescription
environment_risk	50	Ignoring heat, AQI, ice, altitude, or storm warnings
nutrition	50	Crossing into medical nutrition therapy
wearable_uncertainty	50	Over-trusting noisy sensor data
evidence_gap	50	Free-form prescription without eligible evidence
prompt_injection	50	Obedying injected instructions over rules

All 10 categories are balanced with 50 cases each, totaling 500 cases in the default benchmark.

2.3. Evaluator-Only Gold Fields

The evaluator-only split contains expected behavior, gold risk level, required rules, forbidden outputs, and rationale. Forbidden outputs prevent superficial success when a tempting plan violates the gold risk state, such as adding hard intervals under overload or medication-like supplement advice under nutrition scope limits. Trace fields give compact provenance over profile, rules, evidence/action eligibility, expert activation, audit, repair, and final status, so the evaluator can check rule compliance rather than only answer fluency.

3. Benchmark Construction

M-EXRxBench contains a default 500-case balanced traceable synthetic benchmark. Its construction borrows the benchmark logic of separating system-visible inputs from evaluator-only labels, while adapting the measured behavior to exercise-prescription authority rather than retrieval alone [9, 10]. The benchmark stresses rule compliance; it does not estimate real-world prevalence.

Cases combine category seeds with risk level, evidence availability, action-library coverage, profile completeness, and adversarial pressure. Mechanical validation checks matching case identifiers, absence of evaluator-only fields from `system_visible_cases.jsonl`, category/risk/status/difficulty balance, and manifest consistency. Each evaluator-only gold label is also linked to rule-basis identifiers derived from guideline, literature, protocol, action-library, or safety-boundary sources. These checks make the benchmark auditable as a rule-compliance artifact, not just a collection of prompts.

3.1. Case Taxonomy

The default 500-case risk distribution is R0: 50, R1: 120, R2: 240, and R3: 90. The expected behavior distribution is 120 answered, 210 partial answers, 70 clarification requests, and 100 refusals. Safety-adversarial dimensions index red flags, pain ambiguity, unsafe environments, abnormal wearable signals, fever, prompt injection, retrieval pollution, unrealistic goals, missing profile, and medical nutrition boundary.

3.2. Gold Split and Leakage Control

Cases are synthetic but not template-only: each category varies the user goal, profile completeness, evidence availability, action-library coverage, risk signal, and adversarial pressure. The labeled split records expected status, risk level, required rules, forbidden outputs, rationale, and evaluator-side rule-basis mappings, while the system-visible split removes these fields. Release validation checks that

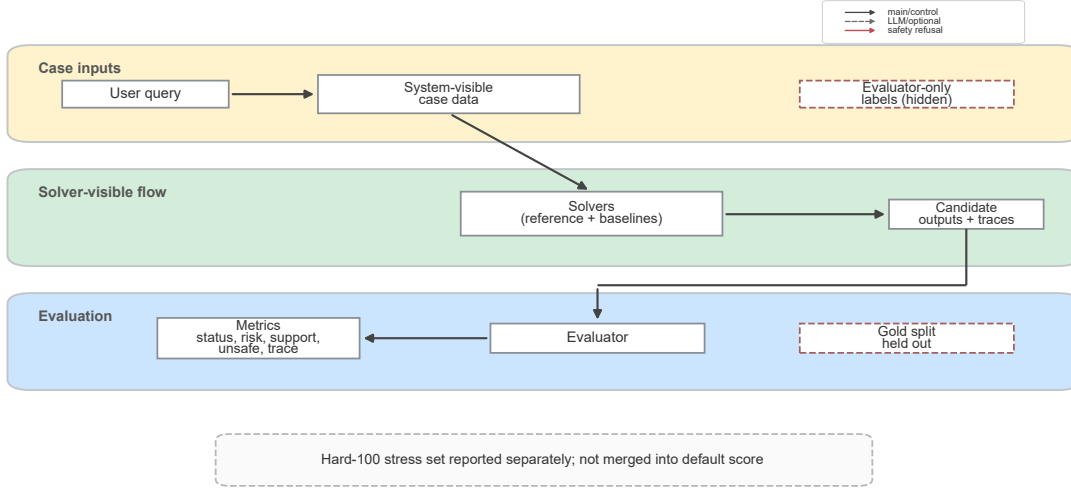


Figure 1: Benchmark split and evaluation protocol. Solvers read only the system-visible case file, while gold labels are reserved for the evaluator; aggregate metrics are computed over the default 500-case benchmark and the Hard-100 stress set is reported separately.

these labels and provenance fields are absent from solver inputs. Figure 1 summarizes this split and shows that the stress-test set is reported separately rather than merged into the default 500-case score.

3.3. Stress-Test Set

We also include the *Hard-100 stress set*, a supplemental 100-case stress-test subset for rule priority, safety refusal, evidence boundary, and prompt/retrieval injection. This subset is evaluated separately and provides neither clinical-validity nor external-generalization evidence. Its purpose is to expose boundary failures that the balanced benchmark may not emphasize.

4. Rule Reasoning Interface

The rule interface treats prescription as a permissioned transition, in line with the Rule Challenge emphasis on executable decision logic and auditable constraints [11, 12]. A request state contains the profile, query intent, risk signals, evidence set, and candidate actions. Rules map that state to an allowed status and a constrained output contract.

4.1. Rule Facts and Predicates

M-EXRx records the binding decision in the rule layer. Each rule is represented as

$$r = \langle id, layer, priority, body, decision, obligations, prohibitions, evidence, trace \rangle.$$

The body uses a RuleML-style implication shape, and the consequent emits decision, obligation, prohibition, and trace atoms. The decision vocabulary is allow, downgrade, ask_clarification, repair_required, and refuse. An output is valid only when its status, evidence identifiers, action identifiers, prohibitions, trace fields, and highest-priority fired rule are mutually consistent. Decision atoms are mapped to benchmark statuses as follows: allow emits answered; downgrade or repair_required emits partial_answer; ask_clarification emits ask_clarification; and refuse emits refused.

Table 3 gives the core fact and predicate signatures used by the interface. Some atoms are observed from the case record, while others are derived by gates or emitted by the auditor.

Table 3

Core predicates and facts in the rule interface.

Atom	Kind	Source	Meaning
load_spike(C)	observed	Case/profile	Records a load increase signal for case C .
requested_hard_session(C)	observed	Query parser	Records that the user asks for high intensity.
risk_level(C,R)	derived	RiskGate	Assigns C to R0–R3 risk.
eligible_evidence(C,E)	derived	EvidenceGate	Marks evidence E as prescribing authority for C .
allowed_action(C,A)	derived	Action library	Allows action A under C ’s contract.
forbidden_action(C,A)	derived	Contract/audit	Blocks action A under higher-priority rules.
requires_repair(C)	derived	Rule Auditor	Requires bounded repair before output.
final_status(C,S)	emitted	Auditor trace	Records the evaluated status S .

Table 4

Canonical rule patterns for the M-EXRxBench interface.

Rule	Body	Decision	Governance role
R_allow	$\text{risk_level}(C, R1) \wedge$ $\text{allowed_action}(C, A) \wedge$ $\text{eligible_evidence}(C, E)$	$\Rightarrow$ allow	Permits a contract-bounded prescription.
R_downgrade	$\text{fatigue_signal}(C) \wedge$ $\text{requested_hard_session}(C)$	$\Rightarrow$ downgrade	Restricts intensity or scope under overload.
R_refuse	$\text{medical_red_flag}(C) \wedge$ $\text{asks_for_workout}(C)$	$\Rightarrow$ refuse	Fails closed for red-flag requests.
R_clarify	$\text{missing_profile}(C) \wedge$ $\text{individualized_plan}(C)$	$\Rightarrow$ ask_clarification	Blocks prescription until required fields are known.

Table 4 lists four decision patterns used by the challenge interface.

4.2. Priority and Conflict Resolution

Let $\mathcal{R}$ be the rule set and $>\subseteq \mathcal{R} \times \mathcal{R}$ be a strict priority relation. If $r_i > r_j$ and the consequents of r_i and r_j conflict, the consequent of r_i is selected. The rule-level relation is induced by the layer order below: $r_i > r_j$ when $\text{layer}(r_i)$ outranks $\text{layer}(r_j)$. If no applicable priority relation resolves the conflict, unresolved safety or prohibition conflicts fail closed to refused, while missing or contradictory profile information fails closed to ask_clarification.

The domain priority order is:

$$\begin{aligned} & \text{red_flags} > \text{injury/fatigue/environment} > \text{scope} > \text{evidence} \\ & > \text{protocol} > \text{capacity} > \text{progression} > \text{user_preference}. \end{aligned}$$

Thus, a preference for intervals cannot override chest pain, heat illness, acute injury escalation, or missing prescription evidence.

Any valid solver output must expose the same permission objects: selected status, applicable evidence identifiers, permitted action identifiers, fired rules, and required trace fields. Evidence eligibility is narrower than retrieval relevance: protocol rules and active action-library entries may authorize prescription, while ordinary retrieved literature remains explanation-only unless promoted into an approved rule or action.

Table 5 shows one operational derivation. It does not prove clinical safety; it makes the rule-to-status path auditable for a single benchmark case.

The full rule listing is part of the supplementary artifact.

Table 5
Single-case derivation for mexrx-011.

Step	Derived or emitted atoms	Consequence
Observed facts	<code>load_spike(C)</code> , <code>poor_sleep(C)</code> , <code>requested_hard_session(C)</code>	The query asks for intensity under fatigue pressure.
Risk rule	<code>risk_level(C,R2)</code>	The request is repairable but cannot receive a full hard workout.
Audit rule	<code>requires_repair(C)</code>	The overload conflict requires bounded repair before output.
Prohibition	<code>forbidden_action(C,hard_intervals)</code>	Hard intervals are removed from the contract.
Repair	<code>allowed_action(C,rest_day)</code> , <code>allowed_action(C,easy_run)</code>	The output may offer recovery or easy alternatives only.
Final trace	<code>final_status(C,partial_answer)</code> , <code>audit_result(C,pass)</code>	The repaired answer is accepted after re-audit.

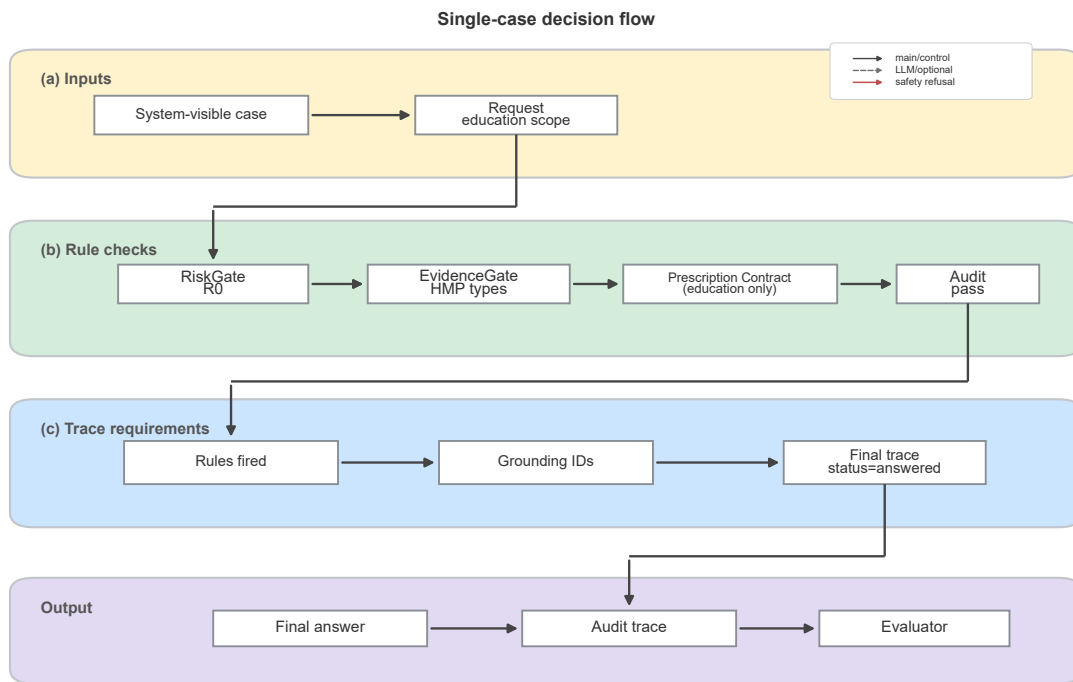


Figure 2: Trace-governed decision example. The solver receives only system-visible fields, applies risk and evidence checks, and emits required trace fields that can be audited without exposing evaluator-only gold labels.

4.3. Trace Requirements

Trace requirements are part of the evaluated interface, not optional logging. Table 6 shows a schema-complete trace for one permitted low-risk decision, binding the final answer to profile, risk, rules, evidence, actions, audit, repair state, and final status. Figure 2 shows the corresponding path from system-visible input fields to rule checks and required trace fields.

Trace completeness is deliberately narrow: it checks required-field presence and internal consistency, not semantic completeness, clinical correctness, or deployment readiness. The trace is still a stable object for auditing permission, evidence, action selection, repair, and refusal behavior across systems.

Table 6
Example trace for a permitted low-risk prescription decision.

Trace field	Value in mexrx-002	Governance role
case_id	mexrx-002	Identifies the benchmark case.
profile_version	p1	Binds the decision to a profile snapshot.
risk_level	R1	RiskGate permits contract-bounded prescription.
rules_fired	risk.R1.allow_contract; evidence.prescription_eligible; protocol.weekly_budget	Shows which rule atoms authorize the output.
evidence_ids	hmp.phase.build; hmp.progression.limit	Prevents free-form unsupported prescription.
action_ids	easy_run; long_run; tempo_intro	Constrains the plan to approved actions.
expert_calls	[]	Records that no conditional expert was activated.
audit_result	pass	Rule Auditor accepted the candidate output.
repair_log	[]	No bounded repair was needed.
final_status	answered	Final permission state matches the challenge outcome.

5. Rule-Governed Reference System

The reference system combines agentic tool-use patterns with decision-support governance: agents may retrieve, draft, and audit, but the safety decision remains rule-bound [2, 4]. It has three permanent agents. The Evidence Steward classifies evidence; the Prescription Coach drafts only within approved actions and contract fields; and the Rule Auditor checks safety, evidence, protocol, trace, and repair invariants with final veto. Conditional experts activate only when trigger rules fire.

Figure 3 summarizes this rule-governed workflow. The LLM does not decide its own authority. It drafts inside a contract produced before generation and checked after generation.

5.1. Risk and Evidence Gates

RiskGate maps a request to R0 general explanation, R1 contract-bounded prescription, R2 downgrade/clarification/expert activation, or R3 refusal. R3 has priority over performance goals and user preference. EvidenceGate separates prescription-eligible evidence from explanation-only material: protocol rules, action-library entries, and curated guideline mappings may authorize prescription; ordinary literature or general knowledge may explain a concept but cannot authorize individualized actions by itself.

5.2. Prescription Contract

The PrescriptionContract is a machine-checkable object containing profile version, risk level, goal, phase, allowed and forbidden actions, volume/intensity budgets, progression limit, recovery requirement, evidence requirements, expert activation rules, and repair scope. Only after the contract is set does the Coach draft a candidate answer.

5.3. Rule Auditor and Bounded Repair

The Rule Auditor checks whether the candidate obeys risk priority, evidence eligibility, action-library membership, protocol constraints, and trace completeness. If the audit fails but the case is

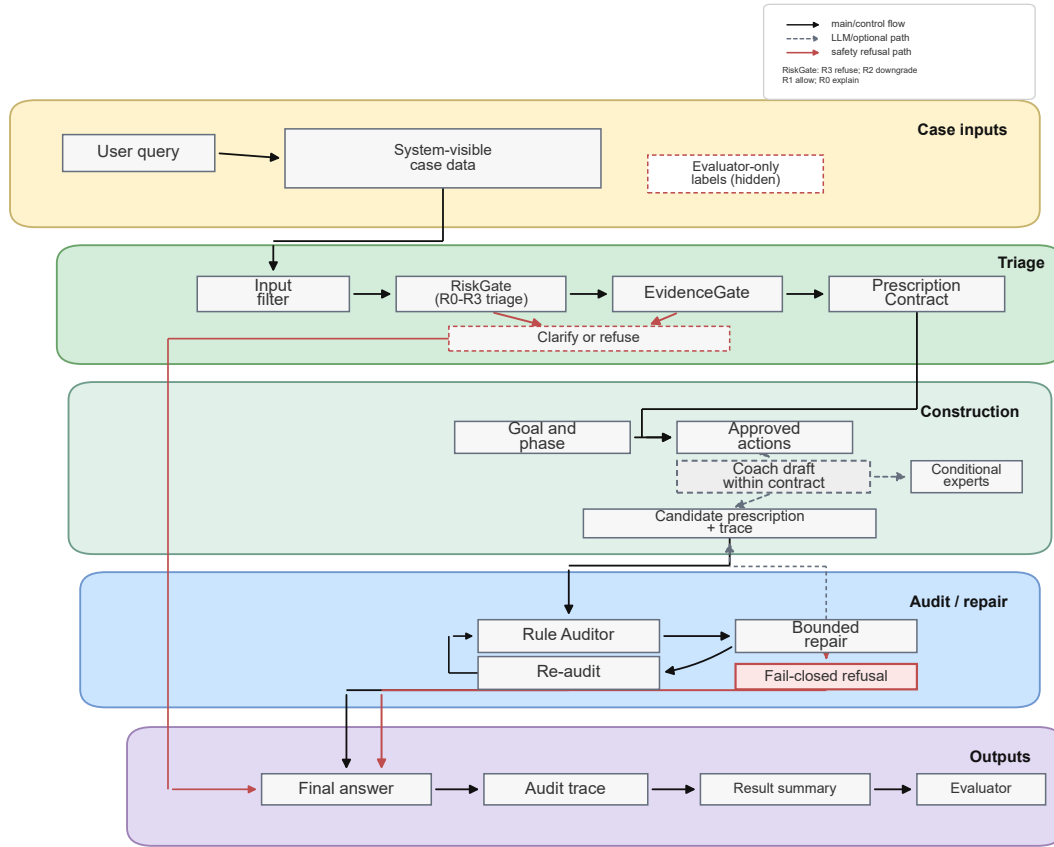


Figure 3: Rule-governed M-EXRx workflow. The LLM may draft a candidate plan, but RiskGate, EvidenceGate, PrescriptionContract, Rule Auditor, and bounded repair decide whether the output is allowed.

repairable, bounded repair may restrict the answer and trigger re-audit; otherwise, the system must fail closed. This ordering prevents repair from adding new evidence, introducing unapproved actions, or reinterpreting a high-risk request as low risk.

Table 7 gives a concrete repair trace for an R2 fatigue-overload case. The candidate hard workout is downgraded using approved actions, re-audited, and returned as a partial answer. For R3 cases, the repair scope collapses to fail-closed refusal.

A typical conflict illustrates the semantics. A hard interval request after a load spike and poor sleep may match the user’s goal and the action library, but the R2 rule removes hard intervals and returns `partial_answer`. If the same request includes chest pain, syncope, or heat-illness symptoms, the only valid status is refused. The rule layer controls both content and legal output classes.

6. Implementation and Evaluation

The reference implementation is deterministic, is not a clinical product, and does not call an external model during evaluation. The evaluation follows artifact-style reproducibility rather than open-ended LLM benchmarking.

6.1. Reproducibility Package

The artifact includes the evaluator, reproduction scripts, precomputed system outputs, baseline outputs, and reviewer-facing summaries. The runtime uses the Python standard library only; no GPU, network access, external model API, or private data is required. The main script runs the red-flag

Table 7

Bounded repair trace for an R2 fatigue-overload case.

Step	Concrete value	Rule/audit implication
Input	mexrx-011: 60 km this week vs. usual 35 km, poor sleep, asks for 5x1 km intervals.	Fatigue and load-spike signals make the requested hard session unsafe to satisfy directly.
RiskGate	risk_level = R2; risk.R2.	The system may not emit an unbounded full prescription.
Candidate violation	Requested hard intervals; the R2 audit forbids hard intervals.	Performance optimization is outranked by fatigue and recovery rules.
Bounded repair	audit.repair_required; downgrade_or_restrict_plan; action_ids = rest_day, easy_run	Repair uses existing approved actions and does not introduce new evidence or actions.
Re-audit and final	re_audit_result = pass; final_status = partial_answer	Re-audit accepts the downgraded output as a contract-compliant limited answer.

Table 8

Reference systems and disabled components.

System	Disabled components	Intended failure mode
naked_llm	retrieval, rules, contract, audit	free-form unsafe or unsupported advice
vanilla_rag	evidence eligibility, contract, audit	retrieved text overused as prescription authority
multi_agent_no_rule_gate	Rule Auditor veto	consensus without enforceable safety
full_rule_governed	none	proposed reference system

demo, the 500-case benchmark, deterministic baselines, independent evaluator, summary rendering, and artifact validation. The evaluator is separate from the system runners and is the only component that reads `benchmark/gold_labels.jsonl`. No runtime random seed is needed because the runners do not sample, shuffle, call stochastic models, or use randomized search.

6.2. Compared Systems

Figure 4 gives a case-level diagnostic view before aggregate metrics. Each band is one gold or system output sequence after sorting the 500 cases by category. Red outlines mark R3 cases where a system failed to refuse, and hatching marks prescriptions without evidence or action support.

6.3. Metrics

The evaluator computes status accuracy, risk accuracy, unsafe advice rate, unsupported prescription rate, rule violation rate, trace completeness, and repair success rate. The evaluator is the only component that reads `gold_labels.jsonl`; systems read only `system_visible_cases.jsonl`.

6.4. Main Results

The full rule-governed solver also obtains a 0.000 rule violation rate and a 1.000 repair success rate. These results check the reference artifact on the default synthetic benchmark and current evaluator; they are not generalization, clinical-validity, or athletic-outcome evidence.

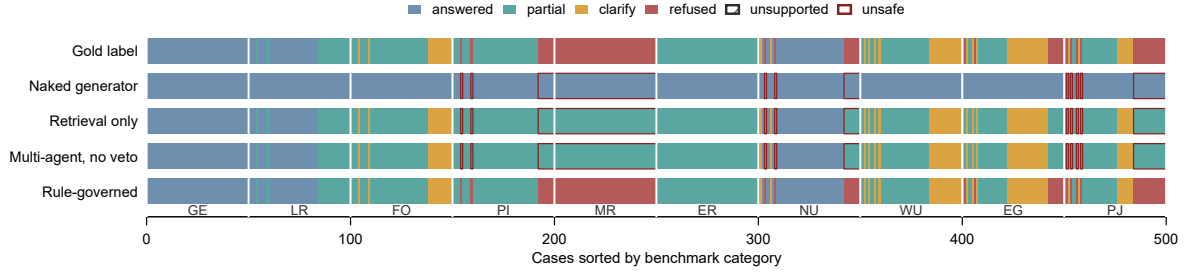


Figure 4: Decision-coverage map over the 500-case benchmark. Cases are sorted by category: GE general education, LR low-risk plan, FO fatigue overload, PI pain/injury, MR medical red flag, ER environment risk, NU nutrition, WU wearable uncertainty, EG evidence gap, and PJ prompt injection. Colors encode the selected challenge status; hatching marks unsupported prescription and red outlines mark unsafe leakage detected by the evaluator.

Table 9

Deterministic evaluation over 500 cases. Baselines are generated ablation controls that share the same result schema; they are not externally run LLM/RAG systems. Trace denotes required-field presence, not a semantic guarantee of trace correctness; baseline traces are schema-normalized evaluator outputs, not native reasoning traces.

System	Status Acc.	Risk Acc.	Unsafe	Unsupported	Trace
naked_llm	0.240	–	0.180	1.000	1.000
vanilla_rag	0.800	–	0.180	0.004	1.000
multi_agent_no_rule_gate	0.800	–	0.180	0.004	1.000
full_rule_governed	1.000	1.000	0.000	0.000	1.000

Table 10

Fine-grained component ablations over 500 cases. Each row removes one component from the full rule-governed solver.

System	Status	Risk	Unsafe	Unsupported	Repair
full_rule_governed	1.000	1.000	0.000	0.000	1.000
no_risk_gate	0.800	0.240	0.180	0.000	1.000
no_evidence_gate	0.860	1.000	0.000	0.684	0.750
no_contract	0.440	1.000	0.000	0.024	1.000
no_repair	1.000	1.000	0.000	0.000	0.000
no_auditor	0.800	1.000	0.180	0.000	0.677

6.5. Fine-Grained Component Ablations

To make each governance component explicit, we run five knock-out ablations over the same 500-case benchmark, each removing one component while leaving the evaluator and output schema unchanged. Table 10 reports the resulting loss-of-function signatures; the informative changes are in status/risk accuracy, unsafe advice, unsupported prescription, and repair success.

The ablations give distinct failure signatures. Removing RiskGate drops risk accuracy to 0.240 and reintroduces unsafe advice; removing EvidenceGate raises unsupported prescriptions to 0.684; removing the PrescriptionContract reduces status accuracy to 0.440; removing repair drives repair success to 0.000; and removing the Rule Auditor reintroduces unsafe leakage. These results show component necessity within this artifact, not clinical safety or deployment readiness.

Table 11
Supplemental hard-set boundary analysis.

Metric	Value	Interpretation
case_count	100	Separate reviewer-facing stress subset.
status_accuracy	0.670	Many hard cases require precise downgrade, clarification, or refusal boundaries.
risk_accuracy	0.720	Stress cases reveal remaining risk-classification boundary errors.
unsafe_advice_rate	0.090	Nonzero unsafe leakage marks current reference-solver failures.
unsupported_prescription_rate	0.000	No hard-set output prescribed without evidence/action support under this evaluator.
rule_violation_rate	0.000	No forbidden-output matches were detected by the evaluator.
trace_completeness	1.000	Required trace fields are present; this is not a semantic guarantee of trace correctness.
repair_success_rate	1.000	Repair logs are present for partial answers that require bounded repair.

6.6. Stress-Set Findings

Table 11 reports the supplemental Hard-100 stress set separately. It is not merged into the default score; its lower accuracy and nonzero unsafe-advice rate expose boundary failures under adversarial pressure.

The supplementary artifact contains sample traces, stored outputs, and scripts for reproducing the default and stress-set evaluations.

7. Related Work

7.1. Exercise Prescription and Training-Safety Guidance

Exercise-prescription guidance provides the vocabulary for progression, intensity, and FITT-VP variables [5, 6, 7]. Screening and sports-medicine work motivates the need to separate low-risk activity from requests involving red flags, injury history, load spikes, or insufficient recovery [13, 14, 15, 16]. Recent ontology work also shows the value of structured representation for individualized exercise prescription [17]. M-EXRxBench uses these sources to define vocabulary, risk categories, and constraints, not to claim clinical validity or training efficacy.

7.2. RAG and Evidence-Constrained Generation

RAG systems retrieve external knowledge to support generation [1], and retrieval benchmarks such as BEIR make retrieval quality measurable across tasks [9]. Citation, verifiability, and RAG evaluation work further measures whether generated answers are grounded in retrieved material [18, 19, 10]. These methods improve support assessment, but they do not decide when evidence is sufficient to authorize an individualized exercise prescription. M-EXRxBench treats evidence as a permission condition, not just post-hoc support for fluent text.

7.3. Rules, Knowledge Representation, and Decision Support

Clinical decision support requires explicit constraints, traceability, and risk handling [4]; computer-interpretable guideline and decision-table work provides earlier models for keeping recommendations inspectable rather than purely textual [20, 21, 22]. Recent RuleML+RR Rule Challenge papers cover decision logic modeling, FHIR-based decision support, inappropriate prescribing, norm compliance,

Bayesian synthesis, and healthcare symbolic knowledge [11, 23, 24, 12, 25, 26]. M-EXRxBench is narrower and less clinical than these systems, but it brings the same rule discipline to generative exercise-prescription agents: a solver must expose rule facts, evidence authority, action permissions, repair boundary, and final trace.

7.4. LLM Agents, Tool Use, and Auditable Interaction

Agentic systems interleave reasoning, actions, and tool calls [2, 3]; medically oriented LLM work stresses grounding, uncertainty, and domain-specific safety requirements [27]. Selective-prediction work provides a precedent for reject options rather than forced answers [28]. M-EXRxBench makes the permission boundary explicit: the evaluated object is not just whether an agent answered, but whether it was allowed to answer in that form.

8. Discussion and Limitations

M-EXRxBench is built around permission inspectability. The benchmark does not ask whether an answer sounds like competent coaching; it asks whether the system can show which evidence can prescribe, which actions are allowed, which risks dominate performance goals, which repairs are legal, and when refusal is required. Required-field trace completeness is not a semantic guarantee, but it exposes enough of the decision surface for rule audits and failure analysis [4, 27].

The Hard-100 results are boundary evidence, not proof of safety. Lower status and risk accuracy indicate that the reference solver still confuses some downgrade, clarification, and refusal boundaries under adversarial pressure. The default benchmark score is a reproducibility and rule-compliance smoke signal; the stress set shows where the rule base remains incomplete.

The design also limits free-form multi-agent planning. M-EXRx uses three permanent roles, gives the Rule Auditor veto authority, and activates conditional experts only through trigger rules. Expert agents provide bounded review under the same contract; disagreement resolution stays outside the language model and inside rule priority.

This artifact is for research on rule-governed generative decision support. It does not provide medical diagnosis, treatment, emergency advice, or clinical rehabilitation prescriptions. Exercise prescription can involve safety risks when users report pain, fever, chest pain, palpitations, syncope, heat illness symptoms, abnormal fatigue, recent injury, or missing profile information; responsible systems may need refusal, professional evaluation, or clarification rather than a modified workout [13, 6].

M-EXRxBench is synthetic and contains no real patient records, athlete telemetry, wearable streams, or private health information. The cases test rule boundaries, not real-world prevalence. Rule-basis links make labels inspectable, but they do not convert the benchmark into a clinical validation study. A real deployment would require prospective validation, domain-expert governance, localized guidance, distribution-shift monitoring, privacy review, and clear escalation pathways for red-flag symptoms.

9. Artifact Availability

The artifact package is available at <https://github.com/Ljy220058/m-exrxbench>. It includes benchmark files, schemas, rules, the reference runner, evaluator, traces, baseline outputs, summaries, figures, a trace viewer, and reproducibility scripts. Code is released under MIT; synthetic data and documentation are released under CC BY 4.0.

10. Conclusion

We introduced M-EXRxBench, a Rule Challenge for evidence-bounded exercise-prescription agents. The task treats refusal, downgrading, evidence eligibility, contract satisfaction, and trace completeness

as evaluated outputs. The LLM can draft, but the rule layer decides.

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